

Class 17 Homework

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```
data <- read.table("rs8067378_ENSG00000172057.6.txt")
head(data)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

Q13.

```
summary(data)
```

sample	geno	exp
Length:462	Length:462	Min. : 6.675
Class :character	Class :character	1st Qu.:20.004
Mode :character	Mode :character	Median :25.116
		Mean :25.640
		3rd Qu.:30.779
		Max. :51.518

```
medians <- aggregate(exp ~ geno, data = data, FUN = median)
medians
```

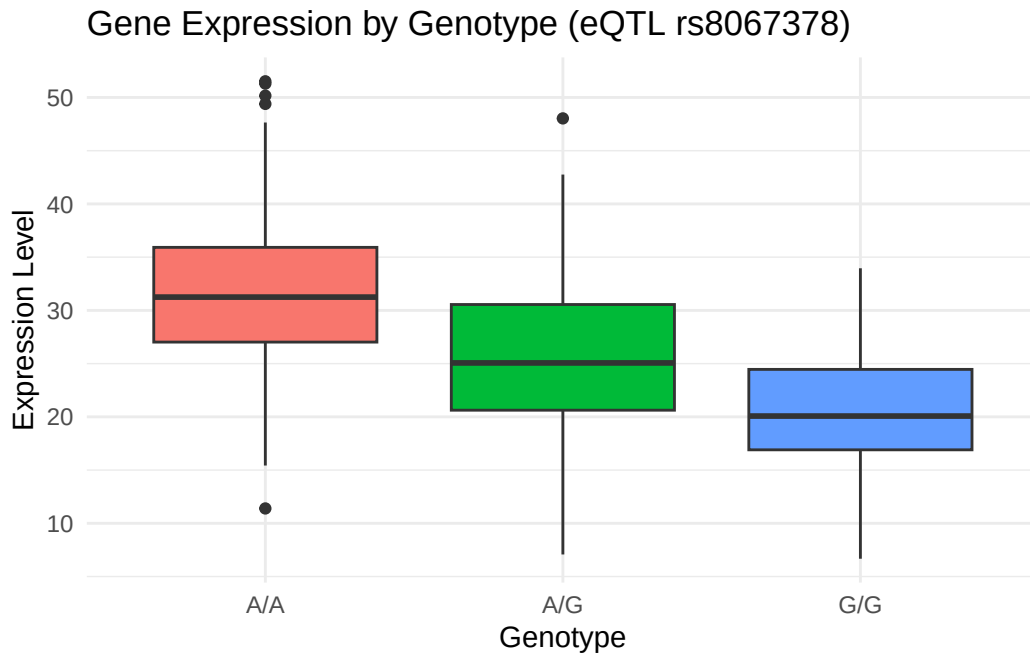
	geno	exp
1	A/A	31.24847
2	A/G	25.06486
3	G/G	20.07363

Q14.

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.2

```
ggplot(data, aes(x = geno, y = exp, fill = geno)) + geom_boxplot() + labs(title = "Gene Expr
```



The plot above suggests that there is reduced expression in the G|G genotype compared to the A|A genotype. Given the distributions of G|G compared to A|A, I would think that there is a reduction in expression of ORMDL3 in G|G individuals, but I have not performed a statistical test that sufficiently shows a difference between the two groups.